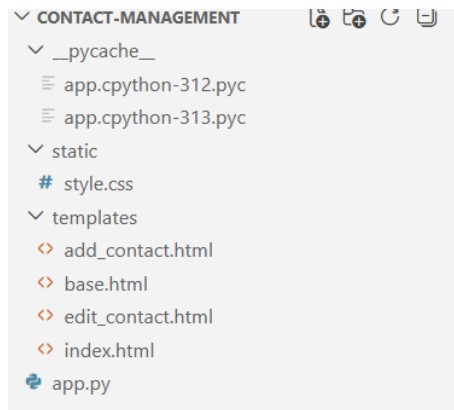


EXPERIMENT – 6 Contact

Management System

#Folder Structure:



#add_contact.html

```
{% extends 'base.html' %}

{% block content %}
<section class="form-shell">
  <div class="form-copy">
    <span class="small-label">Create</span>
    <h1>Add Contact</h1>
    <p>Add complete fake or real contact records with phone, email, city,
company, profile photo, and label.</p>
  </div>

  <form method="POST" class="contact-form">
    <input type="text" name="name" placeholder="Name" required>
    <input type="text" name="phone" placeholder="Phone" pattern="[0-9]{10}"
required>
    <input type="email" name="email" placeholder="Email (@gmail.com)"
pattern=".+@gmail\.com" required>
    <input type="text" name="company" placeholder="Company">
    <input type="text" name="city" placeholder="City">
    <input type="text" name="tag" placeholder="Tag / Role">
    <select name="status">
      <option value="Active">Active</option>
      <option value="Favorite">Favorite</option>
      <option value="Priority">Priority</option>
    </select>
    <input type="text" name="photo" placeholder="Photo URL (optional)">
    <button type="submit">Save Contact</button>
  </form>
</section>
{% endblock %}
```

#base.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Aditya Contact Manager</title>
  <link rel="stylesheet" href="{ url_for('static', filename='style.css')
  }}">
</head>
<body>
  <aside class="sidebar">
    <div class="sidebar-top">
      <span class="small-label">CRM Demo</span>
      <h2>Aditya Contacts</h2>
      <p>Modern contact management system with search, cards, and form
screens.</p>
    </div>
    <nav class="sidebar-nav">
      <a href="{ url_for('index') }}">🏠 Home</a>
      <a href="{ url_for('add_contact') }}">✚ Add Contact</a>
    </nav>
  </aside>

  <main class="main">
    {% with messages = get_flashed_messages(with_categories=true) %}
      {% for category, message in messages %}
        <div class="flash {{ category }}">{{ message }}</div>
      {% endfor %}
    {% endwith %}

    {% block content %}{% endblock %}
  </main>
</body>
</html>
```

#edit_contact.html

```
{% extends 'base.html' %}

{% block content %}
<section class="form-shell">
  <div class="form-copy">
    <span class="small-label">Update</span>
```

```

        <h1>Edit Contact</h1>
        <p>Update contact details while keeping the same modern contact-card
layout from the homepage.</p>
    </div>

    <form method="POST" class="contact-form">
        <input type="text" name="name" value="{{ contact.name }}" required>
        <input type="text" name="phone" value="{{ contact.phone }}"
pattern="[0-9]{10}" required>
        <input type="email" name="email" value="{{ contact.email }}"
pattern=".+@gmail\.com" required>
        <input type="text" name="company" value="{{ contact.company }}">
        <input type="text" name="city" value="{{ contact.city }}">
        <input type="text" name="tag" value="{{ contact.tag }}">
        <select name="status">
            <option value="Active" {% if contact.status == 'Active'
}%selected{% endif %}>Active</option>
            <option value="Favorite" {% if contact.status == 'Favorite'
}%selected{% endif %}>Favorite</option>
            <option value="Priority" {% if contact.status == 'Priority'
}%selected{% endif %}>Priority</option>
        </select>
        <input type="text" name="photo" value="{{ contact.photo }}">
        <button type="submit">Update Contact</button>
    </form>
</section>
{% endblock %}

```

#index.html

```

{% extends 'base.html' %}

{% block content %}
<section class="hero">
    <div>
        <span class="small-label">Dashboard</span>
        <h1>Contact Management System</h1>
        <p>Track classmates, clients, team members, and business connections in
one clean interface.</p>
    </div>
    <div class="hero-stats">
        <div class="stat-box">
            <strong>{{ stats.total }}</strong>
            <span>Total Contacts</span>
        </div>
        <div class="stat-box">
            <strong>{{ stats.favorites }}</strong>

```

```

        <span>Favorites</span>
    </div>
    <div class="stat-box">
        <strong>{{ stats.priority }}</strong>
        <span>Priority</span>
    </div>
</div>
</section>

<form method="GET" class="search-box">
    <input type="text" name="search" value="{{ request.args.get('search', '') }}" placeholder="Search by name, number, company, city, or tag...">
    <button type="submit">Search</button>
</form>

<section class="grid">
    {% for c in contacts %}
        <article class="card">
            <div class="card-accent"></div>
            
            <div class="name">{{ c.name }}</div>
            <span class="badge">{{ c.status }}</span>
            <div class="info">☎ {{ c.phone }}</div>
            <div class="info">✉ {{ c.email }}</div>
            <div class="info">🏢 {{ c.company }}</div>
            <div class="info">📍 {{ c.city }}</div>
            <div class="info">🏷 {{ c.tag }}</div>
            <div class="actions">
                <a href="{{ url_for('edit_contact', contact_id=c.id) }}"
class="btn edit">Edit</a>
                <a href="{{ url_for('delete_contact', contact_id=c.id) }}"
class="btn delete">Delete</a>
            </div>
        </article>
    {% else %}
        <div class="empty-state">
            <h3>No contacts found</h3>
            <p>Try a different search or add a new contact.</p>
        </div>
    {% endfor %}
</section>
{% endblock %}

```

#app.py

```

from flask import Flask, flash, redirect, render_template, request, url_for

```

```
app = Flask(__name__)
app.secret_key = "aditya-contact-manager"
```

```
contacts = [
    {
        "id": 0,
        "name": "Aarav Mehta",
        "phone": "9876543210",
        "email": "aarav.mehta@gmail.com",
        "photo": "https://i.pravatar.cc/150?img=12",
        "company": "PixelForge Studio",
        "city": "Mumbai",
        "tag": "Design Lead",
        "status": "Active",
    },
    {
        "id": 1,
        "name": "Diya Sharma",
        "phone": "9123456780",
        "email": "diya.sharma@gmail.com",
        "photo": "https://i.pravatar.cc/150?img=32",
        "company": "BlueGrid Labs",
        "city": "Pune",
        "tag": "HR Manager",
        "status": "Favorite",
    },
    {
        "id": 2,
        "name": "Kabir Verma",
        "phone": "9988776655",
        "email": "kabir.verma@gmail.com",
        "photo": "https://i.pravatar.cc/150?img=15",
        "company": "Nexa Retail",
        "city": "Delhi",
        "tag": "Client",
        "status": "Active",
    },
    {
        "id": 3,
        "name": "Ananya Roy",
        "phone": "9012345678",
        "email": "ananya.roy@gmail.com",
        "photo": "https://i.pravatar.cc/150?img=47",
        "company": "CodeVista Academy",
        "city": "Kolkata",
    },
]
```

```

        "tag": "Trainer",
        "status": "Favorite",
    },
    {
        "id": 4,
        "name": "Rohan Khanna",
        "phone": "9090901234",
        "email": "rohan.khanna@gmail.com",
        "photo": "https://i.pravatar.cc/150?img=18",
        "company": "ZenCommerce",
        "city": "Bengaluru",
        "tag": "Operations",
        "status": "Active",
    },
    {
        "id": 5,
        "name": "Sneha Iyer",
        "phone": "9345678123",
        "email": "sneha.iyer@gmail.com",
        "photo": "https://i.pravatar.cc/150?img=25",
        "company": "BrightPath Media",
        "city": "Chennai",
        "tag": "Marketing",
        "status": "Priority",
    },
]

```

```

def next_contact_id():
    return max((contact["id"] for contact in contacts), default=-1) + 1

```

```

def find_contact(contact_id):
    return next((contact for contact in contacts if contact["id"] ==
contact_id), None)

```

```

def validate_contact_form(name, phone, email):
    if not name or not phone or not email:
        return "All required fields must be filled."
    if not phone.isdigit() or len(phone) != 10:
        return "Phone number must be exactly 10 digits."
    if not email.endswith("@gmail.com"):
        return "Email must be a valid @gmail.com address."
    return None

```

```

@app.route("/")
def index():

```

```

query = (request.args.get("search") or "").strip().lower()
if query:
    filtered = [
        contact
        for contact in contacts
        if query in contact["name"].lower()
        or query in contact["phone"]
        or query in contact["email"].lower()
        or query in contact["company"].lower()
        or query in contact["city"].lower()
        or query in contact["tag"].lower()
    ]
else:
    filtered = contacts

stats = {
    "total": len(contacts),
    "favorites": sum(1 for c in contacts if c["status"] == "Favorite"),
    "priority": sum(1 for c in contacts if c["status"] == "Priority"),
}
return render_template("index.html", contacts=filtered, search=query,
stats=stats)

```

```

@app.route("/add", methods=["GET", "POST"])
def add_contact():
    if request.method == "POST":
        name = (request.form.get("name") or "").strip()
        phone = (request.form.get("phone") or "").strip()
        email = (request.form.get("email") or "").strip()
        photo = (request.form.get("photo") or "").strip()
        company = (request.form.get("company") or "").strip()
        city = (request.form.get("city") or "").strip()
        tag = (request.form.get("tag") or "").strip()
        status = (request.form.get("status") or "Active").strip()

        error = validate_contact_form(name, phone, email)
        if error:
            flash(error, "danger")
            return render_template("add_contact.html")

        contacts.append(
            {
                "id": next_contact_id(),
                "name": name,
                "phone": phone,
                "email": email,

```

```

        "photo": photo if photo else
"https://via.placeholder.com/100x100.png?text=User",
        "company": company or "Independent",
        "city": city or "Unknown",
        "tag": tag or "Contact",
        "status": status,
    }
)
flash("Contact added successfully!", "success")
return redirect(url_for("index"))

return render_template("add_contact.html")

@app.route("/edit/<int:contact_id>", methods=["GET", "POST"])
def edit_contact(contact_id):
    contact = find_contact(contact_id)
    if not contact:
        flash("Contact not found.", "danger")
        return redirect(url_for("index"))

    if request.method == "POST":
        name = (request.form.get("name") or "").strip()
        phone = (request.form.get("phone") or "").strip()
        email = (request.form.get("email") or "").strip()
        photo = (request.form.get("photo") or "").strip()
        company = (request.form.get("company") or "").strip()
        city = (request.form.get("city") or "").strip()
        tag = (request.form.get("tag") or "").strip()
        status = (request.form.get("status") or "Active").strip()

        error = validate_contact_form(name, phone, email)
        if error:
            flash(error, "danger")
            return render_template("edit_contact.html", contact=contact)

        contact["name"] = name
        contact["phone"] = phone
        contact["email"] = email
        contact["photo"] = photo if photo else
"https://via.placeholder.com/100x100.png?text=User"
        contact["company"] = company or "Independent"
        contact["city"] = city or "Unknown"
        contact["tag"] = tag or "Contact"
        contact["status"] = status
        flash("Contact updated successfully!", "info")
        return redirect(url_for("index"))

```



```

        return render_template("edit_contact.html", contact=contact)

@app.route("/delete/<int:contact_id>")
def delete_contact(contact_id):
    contact = find_contact(contact_id)
    if contact:
        contacts.remove(contact)
        flash("Contact deleted successfully!", "danger")
    else:
        flash("Contact not found.", "danger")
    return redirect(url_for("index"))

if __name__ == "__main__":
    app.run(debug=True)

```

#style.css

```

* {
    box-sizing: border-box;
}

body {
    margin: 0;
    min-height: 100vh;
    font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;
    display: flex;
    background:
        radial-gradient(circle at top left, rgba(26, 188, 156, 0.12),
transparent 18%),
        radial-gradient(circle at bottom right, rgba(52, 152, 219, 0.12),
transparent 20%),
        #ecf3f7;
    color: #21313f;
}

a {
    text-decoration: none;
}

.sidebar {
    width: 260px;
    min-height: 100vh;
    padding: 28px 22px;
    background: linear-gradient(180deg, #253746, #17222c);
    color: #fff;
}

```

```
    position: sticky;
    top: 0;
}

.sidebar-top h2,
.hero h1,
.form-copy h1,
.empty-state h3 {
    margin: 0;
}

.sidebar-top p {
    color: rgba(255, 255, 255, 0.72);
    line-height: 1.6;
}

.small-label {
    display: inline-flex;
    margin-bottom: 12px;
    padding: 6px 10px;
    border-radius: 999px;
    background: rgba(255, 255, 255, 0.12);
    color: #d7f8f1;
    font-size: 12px;
    font-weight: 700;
    text-transform: uppercase;
    letter-spacing: 0.08em;
}

.sidebar-nav {
    display: grid;
    gap: 12px;
    margin-top: 30px;
}

.sidebar-nav a {
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 13px 14px;
    border-radius: 12px;
    color: #fff;
    background: rgba(255, 255, 255, 0.06);
    transition: transform 0.25s ease, background 0.25s ease;
}

.sidebar-nav a:hover {
    background: #1abc9c;
}
```

```
    transform: translateX(8px);
}

.main {
  flex: 1;
  padding: 28px;
}

.flash {
  padding: 13px 16px;
  border-radius: 12px;
  margin-bottom: 14px;
  color: #fff;
  animation: fadeIn 0.45s ease;
}

.success {
  background: #2ecc71;
}

.danger {
  background: #e74c3c;
}

.info {
  background: #3498db;
}

.hero,
.form-shell {
  display: grid;
  grid-template-columns: 1.1fr 0.9fr;
  gap: 24px;
  background: rgba(255, 255, 255, 0.76);
  border: 1px solid rgba(18, 44, 62, 0.08);
  border-radius: 28px;
  padding: 28px;
  box-shadow: 0 18px 45px rgba(20, 48, 68, 0.08);
  backdrop-filter: blur(14px);
}

.hero p,
.form-copy p,
.empty-state p {
  color: #5b6c7a;
  line-height: 1.7;
}
```

```
.hero-stats {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 14px;
  align-self: center;
}

.stat-box {
  padding: 18px;
  border-radius: 20px;
  background: linear-gradient(135deg, #f7ffff, #ecf7ff);
  border: 1px solid rgba(26, 188, 156, 0.18);
  text-align: center;
}

.stat-box strong {
  display: block;
  font-size: 28px;
  color: #1d4f63;
}

.stat-box span {
  color: #5e7483;
  font-size: 14px;
}

.search-box {
  display: flex;
  gap: 12px;
  margin: 22px 0;
}

.search-box input,
.contact-form input,
.contact-form select,
.contact-form button,
.search-box button {
  border-radius: 12px;
  border: 1px solid #cfdbe4;
  padding: 12px 14px;
  font-size: 15px;
}

.search-box input {
  flex: 1;
  background: #fff;
}
```

```

.search-box button,
.contact-form button {
    background: linear-gradient(45deg, #1abc9c, #3498db);
    color: #fff;
    border: none;
    font-weight: 700;
    cursor: pointer;
}

.grid {
    display: grid;
    grid-template-columns: repeat(auto-fill, minmax(255px, 1fr));
    gap: 20px;
}

.card {
    position: relative;
    background: #fff;
    border-radius: 20px;
    padding: 22px 20px;
    text-align: center;
    box-shadow: 0 12px 28px rgba(0, 0, 0, 0.08);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
    overflow: hidden;
}

.card:hover {
    transform: translateY(-8px) scale(1.02);
    box-shadow: 0 18px 36px rgba(0, 0, 0, 0.14);
}

.card-accent {
    position: absolute;
    top: 0;
    left: 0;
    width: 100%;
    height: 5px;
    background: linear-gradient(90deg, #1abc9c, #3498db);
}

.card img {
    width: 92px;
    height: 92px;
    border-radius: 50%;
    object-fit: cover;
    border: 4px solid #1abc9c;
    margin-top: 10px;
    transition: transform 0.25s ease;
}

```

```
}

.card:hover img {
  transform: scale(1.08);
}

.name {
  margin-top: 12px;
  font-size: 21px;
  font-weight: 700;
}

.badge {
  display: inline-block;
  margin-top: 8px;
  padding: 5px 12px;
  border-radius: 999px;
  background: #2ecc71;
  color: #fff;
  font-size: 12px;
  font-weight: 700;
}

.card .info {
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 8px;
  background: #f7fafc;
  color: #334858;
  padding: 9px 12px;
  border-radius: 10px;
  margin: 8px 0;
  font-size: 14px;
}

.actions {
  margin-top: 14px;
}

.btn {
  display: inline-block;
  margin: 5px;
  padding: 9px 14px;
  border-radius: 10px;
  color: #fff;
  font-size: 13px;
  font-weight: 700;
}
```

```

    transition: transform 0.2s ease, opacity 0.2s ease;
}

.btn:hover {
    transform: scale(1.08);
    opacity: 0.94;
}

.edit {
    background: linear-gradient(45deg, #f39c12, #e67e22);
}

.delete {
    background: linear-gradient(45deg, #e74c3c, #c0392b);
}

.form-shell {
    align-items: start;
}

.contact-form {
    display: grid;
    gap: 12px;
    padding: 22px;
    background: #fff;
    border-radius: 20px;
    border: 1px solid #dde7ee;
}

.empty-state {
    grid-column: 1 / -1;
    padding: 32px 22px;
    border-radius: 20px;
    text-align: center;
    background: rgba(255, 255, 255, 0.76);
    border: 1px dashed #bfd0db;
}

@keyframes fadeIn {
    from {
        opacity: 0;
        transform: translateY(-10px);
    }

    to {
        opacity: 1;
        transform: translateY(0);
    }
}

```

```
}

@media (max-width: 980px) {
  body {
    display: block;
  }

  .sidebar {
    width: 100%;
    min-height: auto;
    position: static;
  }

  .hero,
  .form-shell {
    grid-template-columns: 1fr;
  }
}

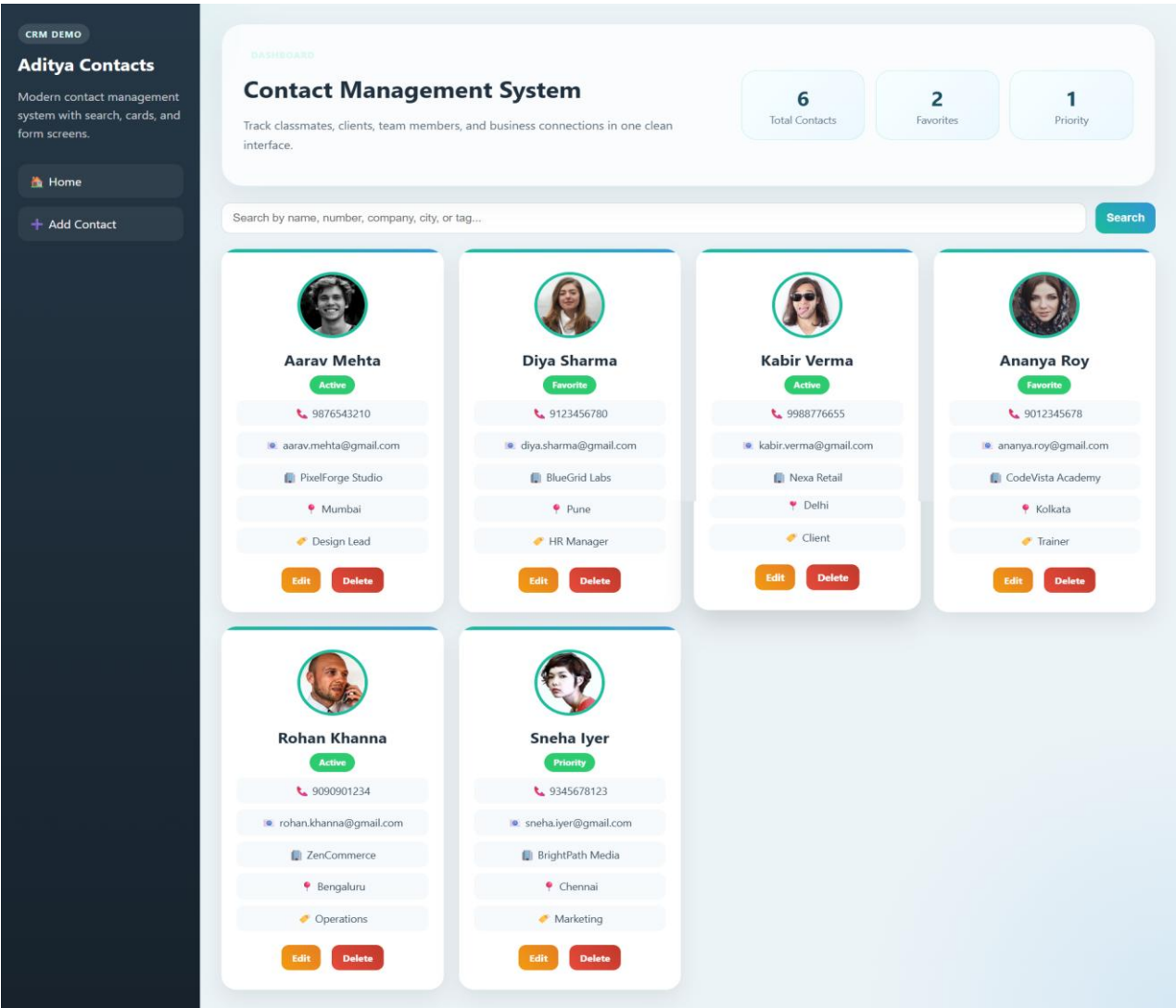
@media (max-width: 620px) {
  .main {
    padding: 18px;
  }

  .hero-stats {
    grid-template-columns: 1fr;
  }

  .search-box {
    flex-direction: column;
  }
}
```


#output:

1. Home Page:



Add Contact Page

CRM DEMO

Aditya Contacts

Modern contact management system with search, cards, and form screens.

Home

+ Add Contact

CREATE

Add Contact

Add complete fake or real contact records with phone, email, city, company, profile photo, and label.

Aditya

8130800696

adiwadhawan@gmail.com

Self

Delhi

Intern

Active

Photo URL (optional)

Save Contact

Edit Contact Page

CRM DEMO

Aditya Contacts

Modern contact management system with search, cards, and form screens.

Home

+ Add Contact

UPDATE

Edit Contact

Update contact details while keeping the same modern contact-card layout from the homepage.

Aditya

8130800696

adiwadhawan@gmail.com

Self

Delhi

Employee

Active

<https://via.placeholder.com/100x100.png?text=User>

Update Contact